Eclampsia

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CABOG.HDOG.MBChB

Eclampsia

- Eclampsia is defined as occurrence of one or more generalized convulsion and or coma in a setting of pre-eclampsia and in the absence of other neurological disorders.

- In the UK it has an estimated surveillance of about 27.5/100,000 pregnancy with case fatality rate estimated to be 3.1%

Eclampsia

- -ante partum (40%),
- - intrapartum (20%), or
- - postpartum (40%).

Eclampsia is obvious as a grand mal convulsion.

However, other causes of fits such as epilepsy have to be considered. Preceding pre-eclampsia suggests eclampsia, but in approximately one-third of cases the eclamptic fit will present first.

- After convulsion the blood pressure will be normal for a while but the protienuria will usually still be present.

- Any convulsion in pregnancy should be considered to be eclamptic until proved otherwise.

- Eclampsia is associated with significant maternal morbidity particularly cerebrovascular event 2.3%.

- Cerebral haemorrhage has been reported to be the most common cause of death in patient with pre eclampsia

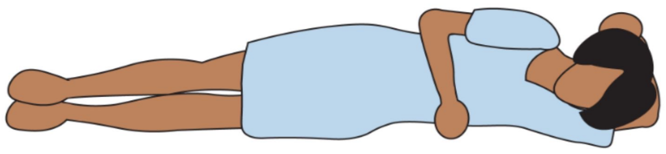
Clinical feature of eclamptic fit is as follow:

- *Premonitory phase*: twitches of small muscles, facial muscles and rolling of eyeballs.
- *Tonic phase*: voluntary muscle is thrown into a state of titanic contractions including intercostals muscles and the diaphragm with respiratory insufficiency and cyanosis.
- *Clonic phase:* muscles forcefully contract and relax; contraction may persist for a minute; they are vigorous and the tongue may be bitten if unprotected.
- *Coma*: occurs with rapid shallow respiration and marked exhaustion.

- ***Following coma:*** patient either passes into another convulsion or regains consciousness.
- Seizures due to eclampsia are usually self-limiting

Management:

- Call for help and approach patient safely, turn her onto her side and try to stop her hurting herself during the fit. Never leave the patient unattended. Bedside rails should be elevated.
- Open airway + check breathing + check circulation.
- Clean airway passage by suctioning and keep patent using a mouth gag and airway.



Left Lateral Recumbent

Call for help

- Auscultate chest and administer oxygen by face mask to correct hypoxia.
- Gain an intravenous access (2 wide bore cannula), send blood for investigation (CBC, renal function test, liver function test, coagulation profile and platelet count, general urine examination including protein urea…etc).
- Insert urinary catheter.

principles

- Anticonvulsant therapy
- Blood pressure control.
- Fluid management
- Delivery
- Postpartum management

Control of fit:

Magnesium sulphate can be used to control an eclamptic fit and to prevent further fit, it act as membrane stabilizer and vasodilator and reduce intracerebral ischemia.

It usually given as a 4g intravenous loading dose and a maintenance infusion of 1-2 gm /hour.

monitor the following:

- Respiratory rate.
- Urine output.
- Patellar reflex.

  Antidote is 10 ml of 10% calcium gluconate.

- Diazepam 10 mg is an alternative but the disadvantage is sedative side effect and need for ventilation.

Control of blood pressure:

- labetalol and hydralazine is the usual first line treatment
- Labetalol: bolus of 20 mg intravenously, followed at 10-minute interval by 40-,80-,80-mg boluses up to 220mg. maintenance is 40 mg /hour.

- Hydralazine: bolus of 5 mg intravenous followed by further boluses of 5 mg up to 15 mg. if still high BP doubling at 30-minute interval or drip of 20 mg in 200 ml normal saline over 30-minute.

If these are in effective other like nifedipine can be used.

- **Asses the fetus**

- Make plane for delivery once stabilized; mode of delivery depend on gestational age and state of cervical dilatation. In most of cases is by C/S.
- Close observation; maintain a fluid chart motoring fluid intake and urinary output via catheter. Chart of vital sign every 30 minute.
- If fit persist an anesthetist is needed urgently to sedate the patient.

Screening and prevention:

- Unfortunately there is currently no screening test for pre-eclampsia. However the ability of the Doppler ultrasound of uterine artery waveform analysis to identify women at risk of pre-eclampsia has been studied.
- In at risk pregnancies there will be characteristic “notch”. This is of value only in high but not in low risk cases with predictive value of 20%.

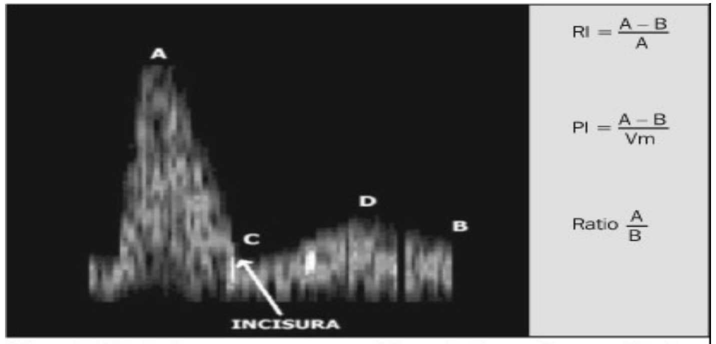
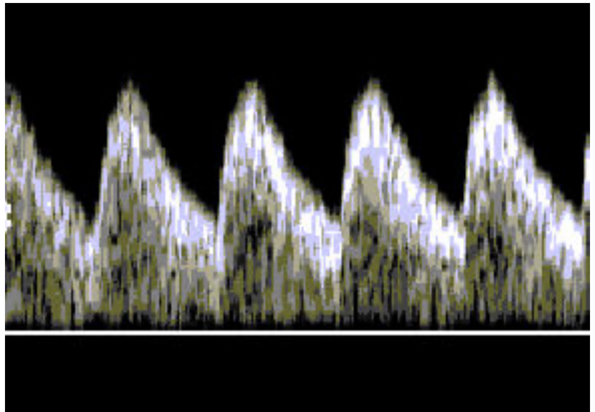


Figure 2. Relation between parameters of flow velocity waveform and the Doppler indexes. (RI, resistance index; PI, pulsatility index; A, systolic peak; B, end-diastole; Vm, mean velocity; C, start of diastole; D, maximum diastole).



Prevention:

Established preventive measures include:

- **Low dose aspirin (75mg) daily, which modestly reduce the risk.**
- **In high risk cases, calcium supplementation may reduce the risk.**
- **Anti oxidant like vitamin C and E, zinc and other are controversial.**

Prevention of pre-eclampsia

Warning signs:

- Epigastric pain, right upper quadrant pain, headache, uncontrolled HT, agitation, hyper-reflexia , poor urine out put, papillodema, facial oedema
- Prevention: low threshold for patient with pre-eclampsia who is unstable or has sever pre-eclampsia, although all patient with pre-eclampsia are at perceived risk of eclampsia regardless of severity

HELLP SYNDROME:

HELLP syndrome is an acronym for haemolysis, elevation of liver enzymes and low platelets.

Women with HELLP syndrome typically present with epigastric pain, nausea and vomiting.

Hypertension may be mild or even absent.

HELLP syndrome is associated with a range of serious complications including acute renal failure, placental abruption and stillbirth.

The management of HELLP syndrome involves stabilizing the mother, correcting any coagulation deficits and assessing the fetus for delivery

HELLP syndrome

- Microangiopathic hemolytic anemia, consumptive thrombocytopenia, liver dysfunction
- 4-12% of patients with severe preeclampsia, 30% occur postpartum
- DIC often secondary to placental abruption, sepsis or fetal death
- Platelet count indirectly proportional to severity of disease
- Complications: ARF, ARDS, hemorrhage, placental abruption, rarely liver hematoma with rupture
- This is a sever form of pre-eclampsia which ma lead to DIC, placental abruption, and fetal death.
- Apart from raised AST and ALT there may be elevated lactate dehydrogenase levels.

- Once dignosed corticosteroid should be administered to enhance lung maturity, also there might be some improvement in HELLP by steroid but final cure is by termination of pregnancy regardless of the gestation

Chronic hypertension

- ***Definition:*** defined as the presence of persistent hypertension, of whatever cause,before the 20th week of pregnancy (in the absence of a hydatidiform mole), or persistent hypertension beyond 6 weeks postpartum.
- This may be manefested for the first time in pregnancy and it may be masked initially by the profound vasodilatation of early pregnancy
- Pre-existing HT will affect 1 -2% of women of reproductive age group. Generally, a sustained blood pressure greater than 140/90 mmHg is deemed hypertensive,but the significant of which depends on clinical situation.

Etiology;

- Overall 95% of HT is known as primary or essential.
- 5% is secondary to other causes mainly renal causes glomerulonephritis, tubointerstitial disease, renal stone or renal artery stenosis. Or it may be secondary to endocrine causes like Cushing's syndrome, Conn's syndrome, phyaeochromocytoma or thyroid disease.

Managements

- ***Preconception, assessment and counseling:***

  Pre-pregnancy counseling should be directed toward the following:

- They should be advised about appropriate anti-hypertensive therapy suitable for pregnancy. Both diuretic and ACE inhibitors should be avoided.

- Life style modification: it's known that high BMI is associated with increased blood pressure, and that weight loss may reduce hypertension. In addition, it has also been shown that both salt and alcohol intake are associated with high BP, and modification of both are advised.
- Proper history taking to look for secondary causes like renal disease which might worsen during pregnancy.

- Detailed physical examination should calculate BMI, fundoscopy for retinal involvement, abdominal examination for palpable kidney as in polycystic kidnys, auscultation over renal artery for possible bruits.

- Women who are found to be hypertensive before pregnancy or early in pregnancy should have circulating level of urea, electrolyte, creatinin and 24-hour urine collection.
- A raised urea and creatinine level may indicate renal impairment and renal causes of hypertension.

Risk facter for developmennt of super-imposed PE

- Renal disease
- Age >40
- Pre-existing DM
- Multiple pregnancy
- APS
- BIM >35
- Previous pre-eclampsia

Antihypertensive drugs:

- Diuretics: it's generally believed that diuretics should be avoided in pregnancy as they acts by increasing renal excretion of sodium and water, which will reduce blood volume and this is not good during pregnancy.

- Angiotensin-converting enzyme inhibitors and ACEI angiotensin receptor antagonists:

  These drugs are associated with renal toxicity in the fetus, and should be avoided in pregnancy.

Antihypertensive that allowed during pregnancy:

- Centrally acting drugs like methyldopa, calcium channel blockers, and labetalol (a combined alpha and beta blocker), have all been used in pregnancy. There is no significant different between these and one can continue this therapy into pregnancy if necessary.

- The drug with the longest, most established safety profile is methyldopa; both neonatal and longer term outcome data have been assessed with no detrimental effect.
- As methyldopa can cause depression and tiredness and very occasionally liver dysfunction, it's not good during postpartum period.

Antenatal

- Once pregnant, the women with underlying HT need close frequent monitoring for the development of pre-eclampsia. More than one in five women with chronic hypertension will develop pre-eclampsia.
- At each visit, BP and urine analysis are checked.
- Uterine artery Doppler velocity waveform has been used to asses risk in these individuals.
- The most common complication of chronic HT is IUGR and second one is placental abruption.
- Frequent monitoring of fetal well being by ultrasound and biophysical profile and CTG is necessary.

Treatment:

- The aim is to maintain diastolic blood pressure below 80mmHg. Mild HT 140/90 may require no treatment but need monitoring.
- Treatment of moderate HT is controversial; however, it may be associated with reduced incidence of development of sever HT.

- Sever HT 160/110mmHg need immediate treatment to prevent risk of stroke and other morbidity.
- Labour is managed as per normal but sever uncontrolled HT may demand premature termination.

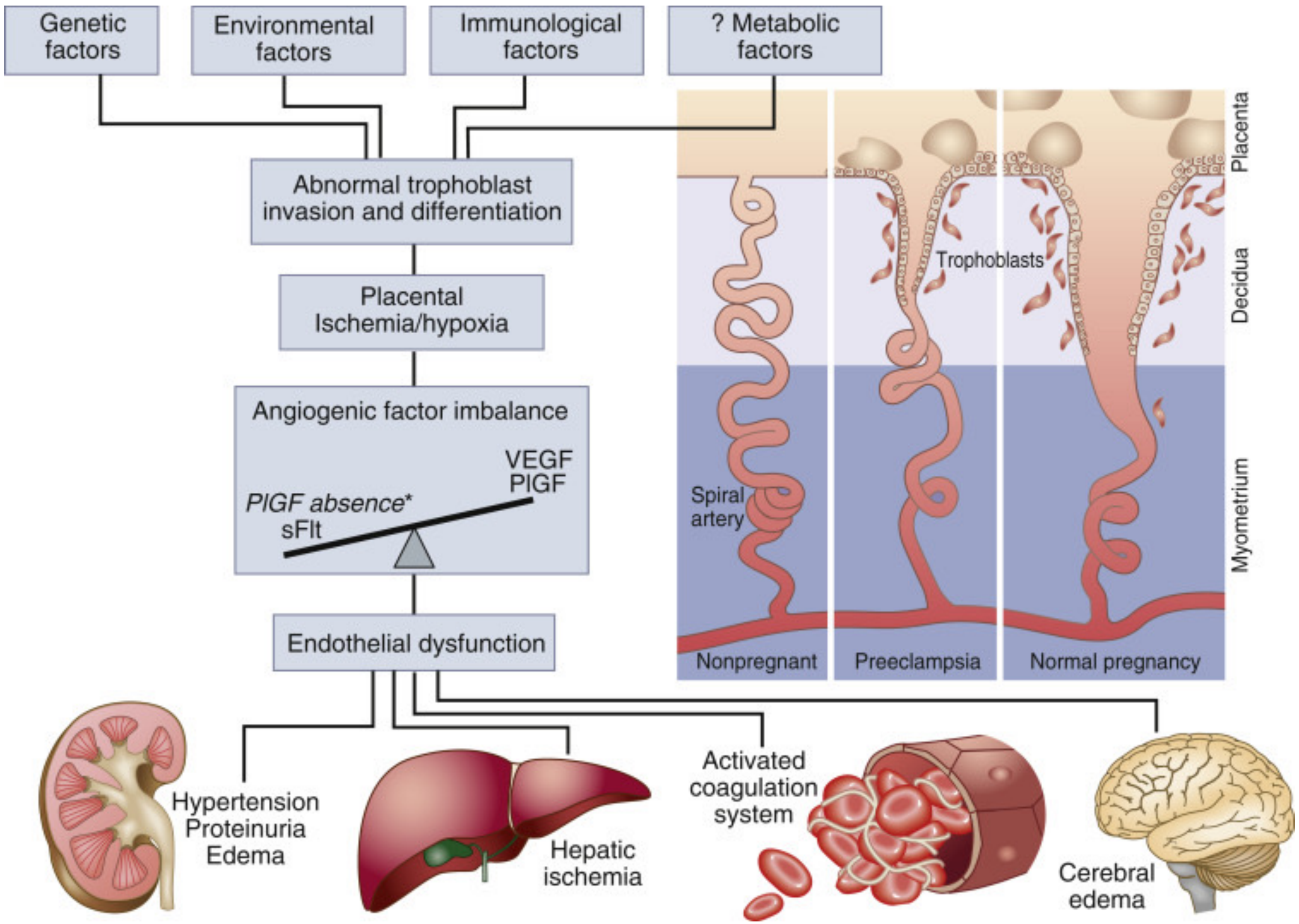
Postpartum:

Women with sever HT need careful monitoring during first 48 hour postpartum as they are at risk of developing renal failure, pulmonary odema and encephalopathy.

Antihypertensive should be continued in the immediate puerperium and then be reviewed 2 weeks later.

Methyldopa should be discontinued and pre-pregnancy drug reintroduced.

THANKS



Genetic factors
Environmental factors
Immunological factors
? Metabolic factors
Abnormal trophoblast invasion and differentiation
Placental Ischemia/hypoxia
Angiogenic factor imbalance
VEGF
PIGF
PIGF absence*
sFlt
Endothelial dysfunction
Placenta
Decidua
Myometrium
Trophoblasts
Spiral artery
Nonpregnant
Preeclampsia
Normal pregnancy
Hypertension
Proteinuria
Edema
Hepatic ischemia
Activated coagulation system
Cerebral edema